AI Models Memorize Gene Names Instead of Learning Pathway Biology

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***Abstract*—Curated pathway databases such as KEGG are widely used as ground truth for judging whether artificial intelligence (AI) systems, including large language models (LLMs), can recover the type of a gene–gene regulatory relation. We show that the standard benchmark design rewards shortcuts. Auditing a public KEGG-derived benchmark of 3,207 relations from 23 human pathways, we find that 19.6% of gene pairs carry more than one label, 41.0% of random-split test rows share their gene pair with training, and gene symbols encode function through family stems. A classifier that reads only the two gene symbols reaches a macro-F1 of 0.731 under the random-row protocol, within the LLM-reported range. We reformulate the task as pair-level multi-label prediction and evaluate seven predictors, including a character-level convolutional network and an inductive graph neural network, under four splits of increasing strictness. Macro AUROC falls from 0.99 when only pairs are held out to 0.81 when gene families are held out and 0.70 when whole pathways are held out (best model, 95% CI 0.66–0.74). The collapse is selective: phosphorylation remains predictable in unseen pathways (AUROC 0.87) because kinase stems are learnable, whereas the activation/inhibition sign falls to 0.56–0.65, although the sign is consistent across pathways for 83% of shared pairs. Scrambling symbol spelling lowers unseen-gene AUROC by 0.09, and conformal coverage falls from 91% to 58% on unseen pathways. High random-split scores measure familiarity with gene names, not pathway knowledge. We release all splits and code, and recommend family- and pathway-disjoint evaluation with symbol-only baselines.**

***Index Terms*—Artificial intelligence, benchmark design, biomedical knowledge mining, data leakage, gene–gene interaction, graph neural networks, KEGG, large language models, shortcut learning.**

I. Introduction

SIGNALING pathways encode which gene products activate, inhibit or phosphorylate one another, and they are a basic substrate for mechanistic reasoning in precision oncology, pharmacology and systems biology. The Kyoto Encyclopedia of Genes and Genomes (KEGG) [1], [2] is among the most widely used manually curated sources of such relations, alongside STRING [3] and OmniPath [4]. Because curation is slow and expensive, there is strong interest in recovering relations automatically, first with biomedical language models [5], [6] and relation-extraction corpora [7], and now with general-purpose large language models (LLMs) [8]–[13].

[Authors: add funding here.]

In this line of work KEGG serves as the answer key: a model receives a gene pair, names the relation type, and is scored against the curated label. High scores are then read as evidence that the model *knows* pathway biology or that a prompting strategy helps it reason. That reading is valid only if the benchmark does not reward shortcuts. For pair-input prediction it often does: when test pairs share members with training pairs, performance is inflated by node-level regularities rather than pair-level understanding [14]–[16]. Related concerns have been raised for gene regulatory network validation [17], and more generally as data leakage [18] and shortcut learning [19].

KEGG relation benchmarks add two complications that have not been examined. First, labels are typed and co-occur: a kinase can both phosphorylate and activate its substrate. Second, genes are named by HGNC symbols [20], whose family stems (*MAPK*, *PRKA*, *DUSP*) carry functional information that any model, including an LLM pre-trained on biomedical text, can exploit without knowing anything about the specific pair.

Here we ask what a KEGG gene–gene relation benchmark actually measures. Fig. 1 summarizes the study. Our contributions are:

- **A structural audit** of a public benchmark released with a recent LLM prompt-optimization study [13], quantifying label co-annotation, cross-pathway redundancy and pair overlap, and the ceiling these impose on single-label accuracy.
- **A shortcut demonstration**: a model that reads only gene symbols matches the performance band reported for LLMs under the random-row protocol.
- **A leakage-aware reformulation** as pair-level multi-label prediction with four splits of increasing strictness, and an evaluation of seven predictors, from a non-learned lookup to a character-level convolutional network (CharCNN) and an inductive graph neural network (GNN).
- **A mechanistic decomposition** showing that pair-disjoint scores are explained by seen genes, that relation mechanism transfers across pathways through kinase-family stems while relation sign does not, and that this gap is not caused by inconsistent sign annotation.
- **Four stress tests**, namely symbol scrambling, direction swapping, calibration under distribution shift and an overlap analysis of held-out pathways, together with model ensembling, precision–recall analysis, split-conformal un-

certainty quantification, curation ranking and a label-permutation control, which confirm that the models rely on name spelling and gene roles and become overconfident exactly where pathway knowledge is required.

All numbers are produced by released code from public data and can be regenerated on a laptop CPU.

II. Related Work

A. AI for Pathway Knowledge

Domain-specific encoders such as BioBERT [5] and PubMedBERT [6] established strong baselines for relation extraction from text, and corpora such as BioRED [7] define typed relations at document level. With LLMs, attention has shifted to whether models can produce pathway relations from parametric knowledge. Evaluations have covered molecular interaction and pathway extraction [9], [10], chatbot-assisted biological pathway curation [11], broad biomedical NLP tasks [8], knowledge extraction from pathway figures [12], and automated prompt search with tree-structured Bayesian optimization [13], [21], and multi-modal LLM–graph models of kinase–substrate–phosphatase regulation [22]. All rely on a trusted reference, typically KEGG.

B. Evaluation Pitfalls for Pair-Input Prediction

Park and Marcotte [14] showed that pair-input predictors tested on pairs whose members appear in training are dominated by component-level effects, and proposed stratifying test pairs by how many members were seen. Hamp and Rost [15] and Bernett *et al.* [16] documented similar inflation in protein interaction prediction, where deep models exploited node degree and sequence similarity. Tabe-Bordbar *et al.* [17] showed that random cross-validation overestimates gene regulatory network accuracy, Kapoor and Narayanan [18] catalogued leakage across machine-learning-based science, and Geirhos *et al.* [19] framed shortcut learning in general.

C. Link Typing in Networks

Predicting the type or sign of an edge from network structure has a long history [23], [24] and has been applied to biomedical graphs, including typed drug–drug edges [25] and protein interactions via path-based similarity [26]. Node embeddings [27] and inductive message passing [28] are standard inputs to such predictors. We use them here as probes of what information in the benchmark suffices to score well.

III. Data and Structural Audit

A. Source

We used the KEGG gene-to-gene relation tables introduced in [13]: one table per KEGG human pathway listing a starter (source) symbol, a receiver (target) symbol and a relation name. The 23 tables cover MAPK, ErbB, Ras, Rap1, cGMP–PKG, cAMP, mTOR, AMPK, p53, apoptosis (two maps), necroptosis, autophagy, endocytosis, longevity, oocyte meiosis, cytokine and chemokine signaling, endoplasmic-reticulum protein processing, adrenergic signaling in cardiomyocytes, and three drug-resistance pathways (EGFR tyrosine kinase inhibitor, endocrine and platinum). Although stored with a `.csv` extension, the files are spreadsheets; we read all 23 and concatenated them with a pathway tag.

TABLE I
Structural Audit of the KEGG-Derived Benchmark

Quantity	Value
Records (all types) / retained	3,207 / 3,166
Pathway tables / distinct genes	23 / 860
Exact duplicate records within a pathway	192
Duplicate (starter, receiver, relation) triples	557
Unique directed gene pairs	2,176
Pairs with > 1 label	426 (19.6%)
activation + phosphorylation	211
inhibition + phosphorylation	204
other combinations	11
Pairs in > 1 pathway (max.)	244 (10)
Gene families (suffix rule)	432
Random 70/20/10 row split, 20 seeds	
Test rows whose pair is in training	$41.0 \pm 1.6\%$
Test rows whose exact triple is in training	$22.9 \pm 1.1\%$
Upper bound on single-label accuracy	0.842

B. Composition

The combined data contain 3,207 records (Table I, Fig. 2a). Activation (1,426), inhibition (903) and phosphorylation (837) account for 98.7%; the remaining 41 records of five rare types were excluded, leaving 3,166 records over 860 genes. Label mixes differ sharply between pathways: the two cytokine maps contain only activation, AMPK and oocyte meiosis are dominated by phosphorylation, and apoptosis by inhibition.

C. Label Co-Annotation, Redundancy and Overlap

Collapsing records to unique directed pairs shows that 426 of 2,176 pairs (19.6%) carry more than one label (Fig. 2b), nearly always phosphorylation together with a sign. This is biologically expected, because phosphorylation describes the *mechanism* of an interaction and activation or inhibition its *effect*, and KEGG annotates both. Treating records as independent single-label instances therefore makes some answers ambiguous by construction; a predictor forced to output one label per pair can reach at most 0.842 accuracy. In addition, 244 pairs occur in several pathways (up to ten) and 192 records are exact within-pathway duplicates. Under a random 70/20/10 row split, 41.0% of test rows share their gene pair with training and 22.9% share the identical (pair, label) triple.

IV. Methods

A. Task Formulations

Task R (row-level, single-label) reproduces current practice: each record is an instance with one of three labels, split at random 70/20/10 into training, test and an unused validation set, as in [13]. We also report a pair-disjoint variant in which all records of an unordered pair fall into one partition. Each setting is repeated with 10 seeds.

Task M (pair-level, multi-label) is our proposed formulation: for each unique directed pair (s, r), predict $\mathbf{y} \in \{0, 1\}^3$



Fig. 1. Study overview. (a) Twenty-three curated KEGG human signalling maps provide 3,207 typed, directed relations among 860 genes (schematic; arrow, activation; bar, inhibition; dashed arrow, phosphorylation). (b) The audit reveals four properties hidden by random row splits: co-annotated pairs, test rows whose pair was already seen in training, informative gene-name families, and a ceiling on single-label accuracy. (c) Four leakage-aware splits of increasing strictness (red blocks). (d) Seven predictors read either the two gene symbols or the training graph and output three label probabilities per pair (bars are illustrative). (e) Stress tests, measured values across pair (P), gene (G), family (F) and pathway (Pw) splits: LR-Lex AUROC with original versus scrambled symbols and original versus swapped gene roles, expected calibration error of LR-Lex and GBDT-Graph, and coverage (%) of 90% split-conformal prediction sets. (f) Per-label AUROC of the symbol-only LR-Lex model: phosphorylation transfers to unseen pathways, whereas activation and inhibition fall toward chance.

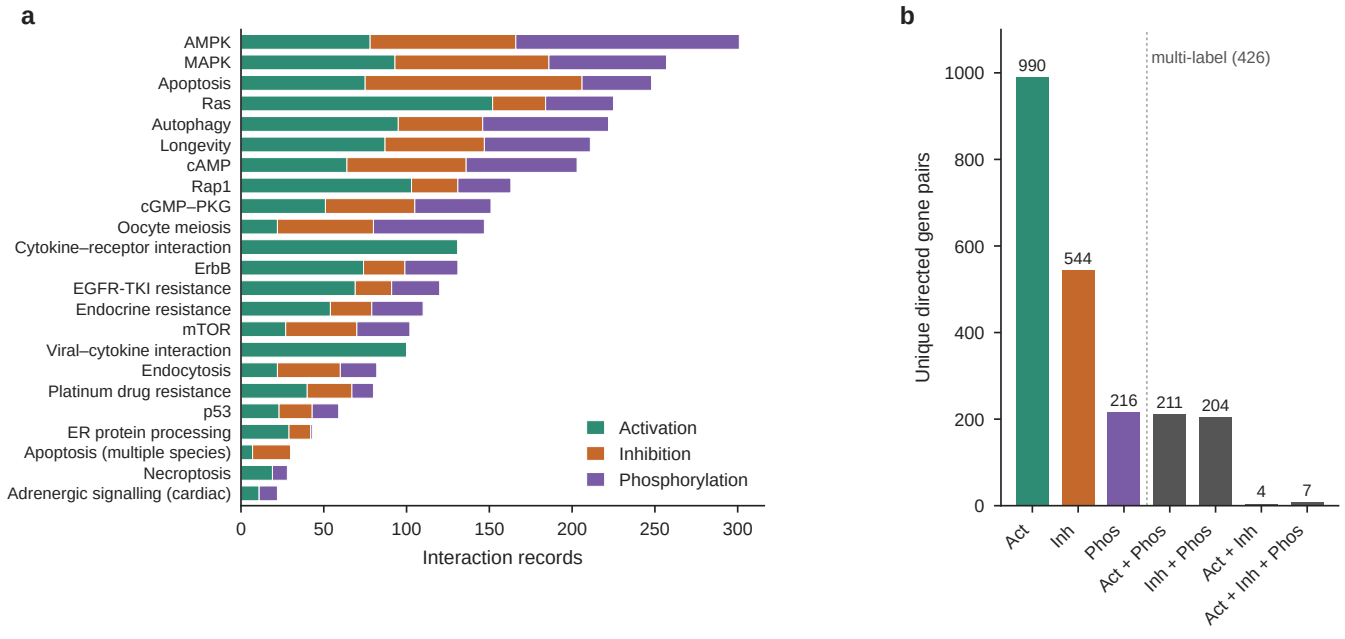


Fig. 2. Benchmark composition. (a) Relation records per KEGG pathway, coloured by type; composition differs strongly between pathways. (b) Label sets of the 2,176 unique directed gene pairs. Coloured bars are single-label pairs; dark bars to the right of the dotted line are the 426 multi-label pairs, almost all of which combine a sign with phosphorylation.

indicating activation, inhibition and phosphorylation [29]. This removes duplicates, represents mechanism and sign explicitly, and makes splits unambiguous.

B. Splitting Regimes

Task M is evaluated under four regimes (Fig. 1c). All features and model parameters are learned from training pairs only.

1) *Pair-disjoint*: unordered pairs are assigned to five folds (two repetitions), so a pair and its reverse never straddle the split.
2) *Gene-disjoint*: genes are assigned to five folds (two repetitions); test pairs involve at least one held-out gene and training pairs none, corresponding to the C2/C3 classes of [14].
3) *Family-disjoint*: as above, but folds are formed over gene families obtained by stripping the trailing numeric suffix of each symbol (e.g., *MAPK1*, *MAPK3* $\rightarrow$ *MAPK*), giving 432 families (largest: CCL, 27 genes). This removes paralog shortcuts.
4) *Pathway-held-out*: each of the 23 pathways is held out in turn; the test set is every pair annotated in it, and training excludes all such pairs in either orientation.

C. Predictors

Prior predicts training label prevalence. *Lookup* is a non-learned node-profile heuristic: the score for label ℓ averages the smoothed frequency $\hat{p}_\ell(g) = (n_{g\ell} + \pi_\ell)/(n_g + 1)$ with which s carries ℓ as a starter and r as a receiver in training, where π_ℓ is the label prevalence.

LR-Lex encodes the two symbols as TF–IDF-weighted character 2–4-grams in separate vocabularies and fits one class-balanced logistic regression per label ($C = 1$). It sees no text, prompt or pathway context.

LR-Graph and *GBDT-Graph* use 48 features of the directed training graph: smoothed starter and receiver label profiles with support counts; in-degree, out-degree and PageRank [30] of both genes; the number of shared intermediates; and a rank-16 truncated SVD embedding of the adjacency matrix with the starter–receiver inner product. They are fitted by logistic regression or by histogram gradient boosting [31] (300 iterations, learning rate 0.05, 15 leaves). *LR-Lex+Graph* concatenates both feature sets.

CharCNN is a neural symbol model in the spirit of [32]: each symbol is embedded character by character (24 dimensions), passed through parallel 1-D convolutions of width 2, 3 and 4 (64 filters each) with max pooling, separately for starter and receiver, and the two codes are combined by a two-layer perceptron with dropout 0.3.

GNN is an inductive two-layer GraphSAGE network [28] with mean aggregation (64 hidden units) over the undirected training graph. Node inputs are a 64-dimensional SVD projection of each symbol's character n-grams plus log degree, so genes absent from training still receive representations. A pair is scored by a perceptron on $[\mathbf{h}_s, \mathbf{h}_r, \mathbf{h}_s \odot \mathbf{h}_r, |\mathbf{h}_s - \mathbf{h}_r|]$.

Both neural models are trained full-batch with AdamW [33] (learning rate 3×10^{-3} and 5×10^{-3}, weight decay 10^{-4}, 250 and 300 epochs) on a class-weighted binary cross-entropy, with fixed hyperparameters across all regimes so that no test information is used for tuning.

For Task R we use a multinomial LR-Lex ($C = 3$), a majority-class baseline, a starter-lookup that returns the starter's most frequent training label (backing off to the receiver, then the global majority), and a pair-lookup that first returns the most frequent training label of the same pair.

D. Evaluation

For Task R we report macro-F1 and the Matthews correlation coefficient (MCC) [34]. For Task M we report macro-averaged AUROC, which is threshold-free, and macro-F1 (threshold 0.5 for learned models and training prevalence for Prior and Lookup), plus per-label AUROC. Prior predicts every label as positive and scores 0.54–0.56 macro-F1, so macro-F1 should be read against it; precision–recall analysis [35] is possible from the released predictions. Pathway-held-out results are test-size-weighted means over pathways, with 95% confidence intervals from 5,000 bootstrap resamples of pathways; the two single-label cytokine pathways have undefined AUROC and are excluded from AUROC averages. Paired comparisons across folds use the two-sided Wilcoxon signed-rank test [36].

E. Stress Tests

Four additional analyses probe what the models rely on, using the same folds. *Symbol scrambling* replaces every gene symbol by a random six-character token (one fixed token per gene, so identity is preserved but spelling is destroyed) and retrains LR-Lex. *Direction swap* scores each test pair with starter and receiver exchanged and compares the predictions with the original labels; a model that uses gene roles should lose accuracy. *Calibration* is measured by the expected calibration error (ECE, 10 equal-width bins, averaged over labels) [37] and the Brier score. *Overlap* is the fraction of gene occurrences in a held-out pathway that also occur in the training pairs, related to per-pathway AUROC by Spearman correlation.

F. Ensembling, Uncertainty and Ranking

Five further analyses use the same folds. *Ensemble* averages the probabilities of LR-Lex, LR-Graph and GBDT-Graph [38]. *AUPRC* is the macro-averaged area under the precision–recall curve [35], compared with label prevalence. *Split-conformal prediction* [39], [40] holds out a random 20% of the training pairs for calibration, refits LR-Lex on the rest, and forms per-label prediction sets with nonconformity $1-\hat{p}(y)$ at a nominal 90% coverage; we report empirical coverage on the test fold, which is guaranteed only if test pairs are exchangeable with calibration pairs. *Curation ranking* measures precision among the ten highest-scoring test pairs per label (precision@10), divided by the label's prevalence in that fold. A *label-permutation control* [41] retrains LR-Lex after shuffling the label vectors of the training pairs; any score above 0.5 would indicate leakage in the pipeline itself. Experiments used scikit-learn [42], NetworkX [43], SciPy [44] and PyTorch [45] on a CPU.

TABLE II
TASK R: ROW-LEVEL SINGLE-LABEL RESULTS (MEAN ± S.D., 10 SEEDS)

Split	Model	Macro-F1	MCC
Random row	Majority	0.208 ± 0.008	0.000
	Pair-lookup	0.628 ± 0.014	0.466 ± 0.022
	Starter-lookup	0.691 ± 0.016	0.558 ± 0.021
	LR-Lex	**0.731** ± 0.011	**0.614** ± 0.016
Pair-disjoint	Majority	0.208 ± 0.004	0.000
	Starter-lookup	0.699 ± 0.026	0.575 ± 0.042
	LR-Lex	**0.780** ± 0.009	**0.688** ± 0.014

TABLE III
TASK M: MACRO AUROC / MACRO-F1 UNDER FOUR SPLITS. MEAN OVER FOLDS (PATHWAY: TEST-SIZE-WEIGHTED MEAN OVER 23 PATHWAYS). BEST AUROC PER COLUMN IN BOLD.

Model	Pair	Gene	Family	Pathway
Prior	.500/.562	.500/.560	.500/.561	.500/.540
Lookup	.974/.863	.891/.738	.785/.652	.597/.484
LR-Lex	.978/.909	**.939**/.844	**.811**/.688	.683/.509
LR-Graph	.982/.922	.871/.686	.763/.555	.620/.447
LR-Lex+Graph	.983/.926	.878/.691	.770/.559	.609/.445
GBDT-Graph	**.987**/.931	.878/.712	.776/.600	.606/.471
CharCNN	.986/.949	.929/.838	.771/.651	**.697**/.520
GNN	**.987**/.948	.895/.772	.779/.631	.674/.510
Test pairs / fold	435	784	769	13–248

V. RESULTS

A. Gene Symbols Alone Match Reported LLM Performance

Under the random-row protocol, LR-Lex reaches a macro-F1 of 0.731 and an MCC of 0.614 from gene symbols alone (Table II). The study that released this benchmark reported macro-F1 between about 0.55 and 0.80 for five LLMs under the same row-level protocol [13]. We did not rerun those models and the samples differ, so the numbers are not directly comparable, but the overlap shows that the protocol cannot by itself separate biological knowledge from symbol-level regularities.

LR-Lex scores *higher* under the pair-disjoint split (0.780) than under the random split. A per-row diagnosis explains why: across 10 seeds, 58.8% of random-split test rows have an unseen pair (accuracy 0.886), 22.7% have a training copy with the same label (0.819), and 18.6% have a training copy with a *different* label (0.209). In the last group the model reproduces the training copy and is marked wrong because the pair is co-annotated. Random-row splitting therefore inflates scores through overlap and deflates them through label conflict at the same time, and its results cannot be interpreted cleanly in either direction.

B. Performance Collapses as Splits Tighten

Table III and Fig. 3 summarize Task M. When only pairs are held out, every learned model exceeds 0.97 macro AUROC, the neural models reach 0.986–0.987, and even the non-learned Lookup reaches 0.974. When one partner is unseen (gene-disjoint), the graph models fall to 0.87–0.90, whereas the two symbol models hold at 0.93–0.94 because unseen genes usually have seen paralogs. Holding out whole families removes this advantage: all models fall to 0.76–0.81 and the spread between them narrows to within fold variability. Under pathway hold-out every model approaches the floor; the best is CharCNN at 0.697 (95% CI 0.659–0.735), followed by LR-Lex at 0.683 (0.640–0.724) and the GNN at 0.674 (0.622–0.729), and no model exceeds the Prior on macro-F1.

The more expressive AI models do not escape the pattern. CharCNN improves on LR-Lex by 0.008 AUROC when pairs are held out ($p = 0.027$) but is 0.010 and 0.040 worse under gene- and family-disjoint splits (both $p = 0.002$). LR-Lex outperforms GBDT-Graph in every regime except pair-disjoint (gene $+0.062$, family $+0.035$, pathway $+0.068$; all $p \leq 0.002$), while GBDT-Graph is slightly better when pairs are held out (-0.008, $p = 0.002$). Model capacity thus helps exactly where shortcuts are available and does not help where they are removed. Per-pathway results for LR-Lex (Fig. 4) range from 0.44 (adrenergic signaling in cardiomyocytes, 13 pairs) to 0.82 (EGFR-TKI resistance), with a median of 0.68, and exceed chance in 20 of 21 evaluable pathways.

C. Seen Genes, Not Seen Pairs, Drive Pair-Disjoint Scores

Stratifying pair-disjoint test pairs by how many of their genes appear in training [14] (Fig. 5a) shows that most test pairs (on average 372 of 435 per fold) have both genes seen; on these, Lookup already reaches 0.977 AUROC and GBDT-Graph 0.990. Pairs with one seen gene (57 per fold) score 0.96 for all three models, and only about 12 pairs per fold have no seen gene, too few to evaluate. A pair-disjoint score therefore mostly measures how well a model recalls what its two genes did elsewhere. The gene-disjoint learning curve (Fig. 5b) makes the same point from another angle: with only 10% of the training pairs (about 140), LR-Lex already reaches 0.851 AUROC, whereas the graph-based Lookup and LR-Graph need all the data to reach 0.87–0.89.

D. Mechanism Transfers; Sign Does Not

The collapse under pathway hold-out is not uniform across labels (Table IV, Fig. 3b,c). Phosphorylation remains predictable in unseen pathways (AUROC 0.82–0.87 for the symbol and neural models), whereas activation and inhibition fall to 0.56–0.65 for every model. The learned weights explain why (Fig. 5d). Phosphorylation is driven by starter-gene n-grams that mark kinase families, such as the prefixes of *MAPK*/*MAP2K* and *PRK*-family kinases; a model can recognize a kinase by name in any pathway. Inhibition draws on prefixes of deubiquitinases (*USP*) and cathepsins (*CTS*) and on adenylyl-cyclase-like receivers, and activation on growth factors and calmodulin; these are weaker, more pathway-specific cues.

E. Sign Failure Is Not Annotation Noise

A natural objection is that sign is simply inconsistent across pathways, so no model could predict it. The data do not support this. Among the 244 pairs annotated in more than one pathway, the sign is identical in every pathway for 83%

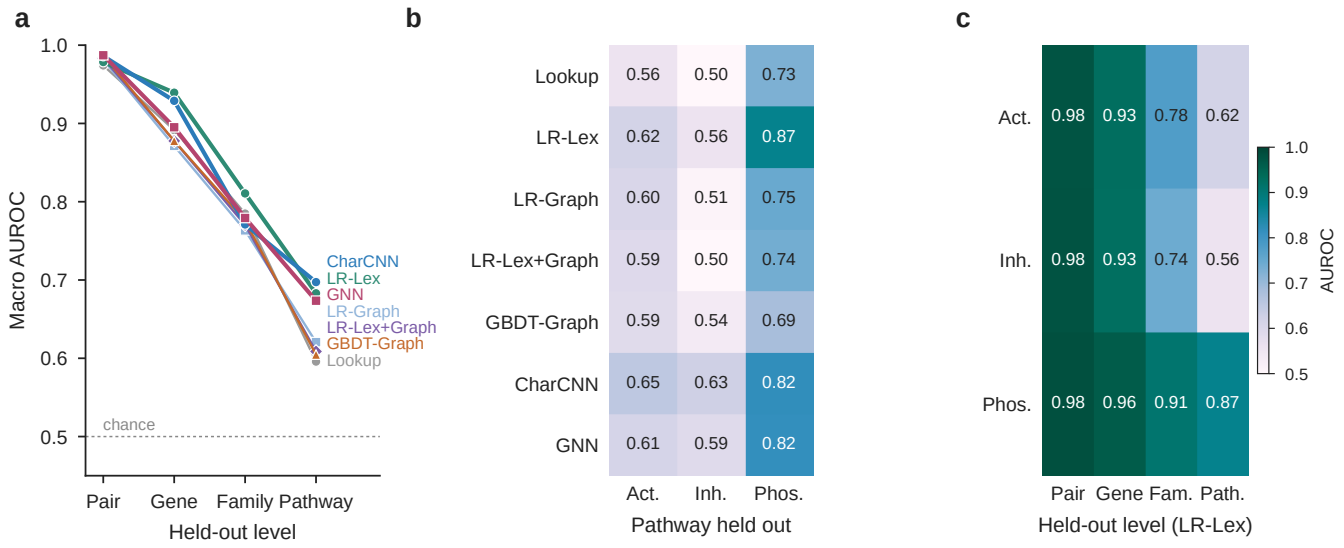


Fig. 3. Performance under four splits of increasing strictness. (a) Macro AUROC of the seven predictors (means over folds; pathway values are test-size-weighted means over pathways); the dashed line marks chance. Fold-level standard deviations are in the released results. (b) Per-label AUROC of every predictor when whole pathways are held out: phosphorylation transfers, whereas activation and inhibition stay near chance for all models. (c) Per-label AUROC of the symbol-only LR-Lex model across held-out levels.

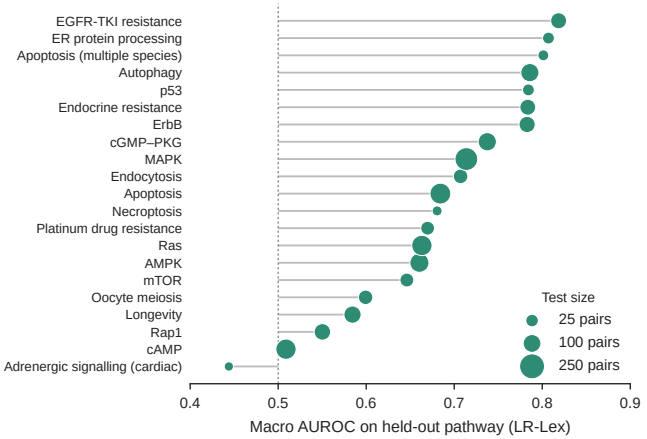


Fig. 4. Macro AUROC of LR-Lex on each held-out pathway (21 evaluable pathways; the two single-label cytokine pathways are omitted). Marker area is proportional to the number of test pairs; the dashed line marks chance.

TABLE IV
PER-LABEL AUROC (ACT. = ACTIVATION, INH. = INHIBITION, PHOS. = PHOSPHORYLATION)

Regime	Model	Act.	Inh.	Phos.
Gene-disjoint	LR-Lex	.930	.926	.962
	CharCNN	.915	.917	.955
	GNN	.879	.882	.924
Family-disjoint	LR-Lex	.779	.744	.909
	CharCNN	.733	.704	.875
	GNN	.755	.714	.869
Pathway-held-out	LR-Lex	.619	.560	.871
	CharCNN	.652	.627	.817
	GNN	.613	.591	.816
	GBDT-Graph	.589	.538	.685

and inhibition status for 95% (Fig. 5c); only 7 of 219 pairs with a sign in several pathways carry opposite signs. The sign of a known pair is therefore largely stable, yet the sign of a *new* pair in an *unseen* pathway cannot be inferred from names or from the topology of other pathways. This is precisely the knowledge a benchmark of pathway understanding should measure, and it is the part that current benchmark designs leave untested.

F. Stress Tests Confirm Reliance on Names and Roles

The stress tests (Fig. 6, Table V) show what the models use. Scrambling symbol spelling barely affects pair-disjoint AUROC (-0.009), where gene identity suffices, but costs 0.089 under gene-disjoint, 0.065 under family-disjoint and 0.094 under pathway hold-out (all $P < 0.01$, Wilcoxon). The lexical model's advantage over graph models on unseen genes (Table III) therefore comes from the spelling of gene names. Swapping starter and receiver lowers pair-disjoint AUROC from 0.978 to 0.655, so the models encode gene roles, such as kinases acting as starters, rather than symmetric similarity.

Calibration degrades with the strictness of the split. GBDT-Graph is the best calibrated model on pair-disjoint data (ECE 0.039 against 0.111 for LR-Lex), yet the worst on held-out pathways (0.373 against 0.235; $P < 10^{-5}$). Its Brier score rises eightfold, from 0.042 to 0.362. Topology-based models are therefore confident precisely where they are wrong, which matters if their probabilities are used to rank candidate interactions for curation.

Finally, difficulty of a held-out pathway is not explained by how many of its genes were seen. Gene overlap ranges from 11% to 79% across pathways, yet its correlation with LR-Lex AUROC is weak and non-significant (Spearman $\rho = 0.15$, $P = 0.52$; GBDT-Graph $\rho = 0.37$, $P = 0.10$; Fig. 6d). Seeing

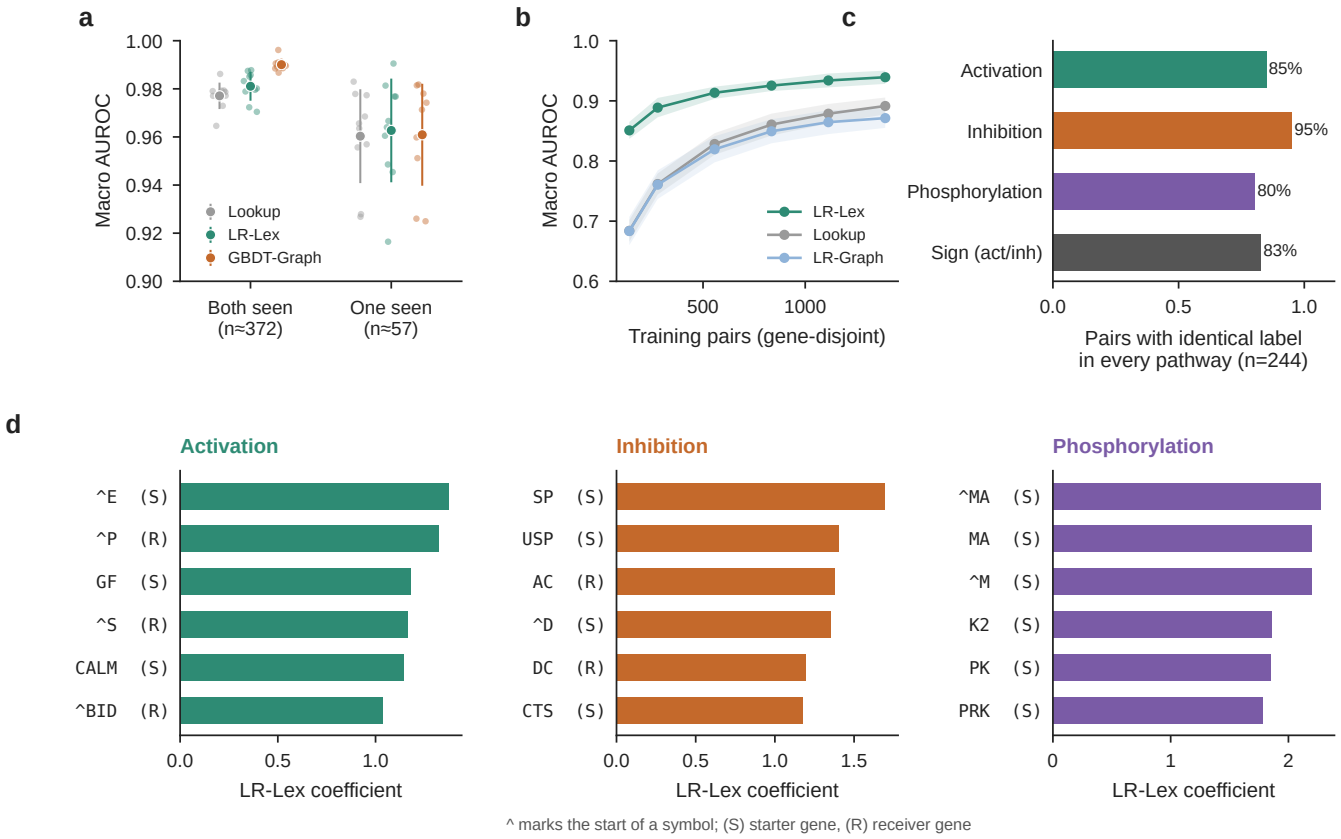


Fig. 5. Where the signal comes from. (a) Pair-disjoint AUROC stratified by the number of genes of each test pair seen in training; small points are folds. (b) Gene-disjoint learning curves; shading, one standard deviation over 10 folds. (c) Fraction of the 244 pairs annotated in more than one pathway whose label is identical in every pathway. (d) The six largest LR-Lex coefficients per label, fitted on all pairs; prefixes of kinase families dominate phosphorylation.

TABLE V
STRESS TESTS FOR LR-LEX AND GBDT-GRAPH (TASK M). AUROC: MACRO AUROC; SCR.: SCRAMBLED SYMBOLS; SWAP: STARTER AND RECEIVER EXCHANGED; ECE: EXPECTED CALIBRATION ERROR. PATHWAY VALUES ARE TEST-SIZE-WEIGHTED.

	Pair	Gene	Family	Pathway
LR-Lex AUROC	0.978	0.939	0.811	0.683
Scr.	0.969	0.850	0.746	0.589
Swap	0.655	0.634	0.590	0.597
GBDT-Graph AUROC	0.986	0.881	0.775	0.608
LR-Lex ECE	0.111	0.097	0.102	0.235
GBDT-Graph ECE	0.039	0.145	0.226	0.373
LR-Lex Brier	0.065	0.098	0.160	0.222
GBDT-Graph Brier	0.042	0.165	0.239	0.362
Test genes seen	92%	41%	36%	55%

a gene in other pathways does not reveal how it acts in a new one, which is consistent with the sign result above.

G. Ensembles, Uncertainty and Ranking Do Not Rescue Transfer

Fig. 7 and Table VI extend the analysis to ensembling, uncertainty quantification and practical use. Averaging three models gives the best pair-disjoint AUROC (0.988, $P = 0.002$ against LR-Lex), but no gain where pathway knowledge is needed: it is below LR-Lex on gene-disjoint (0.924 against 0.939, $P = 0.002$) and pathway-held-out data (0.659 against 0.683, $P = 0.05$), and indistinguishable on family-disjoint data ($P = 0.32$). Model diversity therefore amplifies the shortcut rather than replacing it.

Precision–recall analysis confirms the ranking from AUROC. Macro AUPRC of LR-Lex falls from 0.962 to 0.645 on held-out pathways, against a prevalence baseline of 0.43, and the graph models end at 0.60–0.61 (Fig. 7a).

Split-conformal prediction, which promises a coverage of at least 90% under exchangeability, meets this target only on pair-disjoint data (90.6%). Coverage drops to 85.1% for unseen genes, 72.8% for unseen families and 58.0% for unseen pathways (all $P \leq 0.002$ against 90%, Wilcoxon), with individual pathways as low as 26% (Fig. 7b). Uncertainty estimates calibrated on a random subset of the benchmark therefore cannot be trusted for new pathways.

For curation, the ten top-ranked pairs per label are 1.7–3.4 times enriched over prevalence on pair-, gene- and family-disjoint data, but only 1.16 times for activation and 1.30 times for inhibition on held-out pathways, while phosphorylation keeps a 2.1-fold enrichment (Fig. 7c). A curator using these models to prioritise activating or inhibiting interactions in a new pathway would gain little over random order.

The label-permutation control scores 0.49–0.52 AUROC

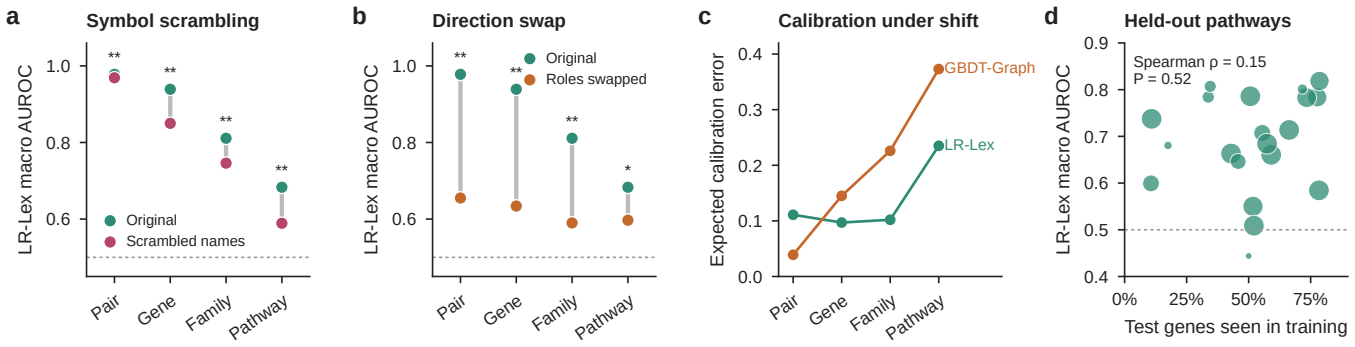


Fig. 6. Stress tests. (a) LR-Lex macro AUROC with original and scrambled gene symbols (identity kept, spelling destroyed). (b) LR-Lex AUROC when starter and receiver are swapped at test time. Asterisks in (a) and (b) give two-sided Wilcoxon signed-rank tests over folds: $*P < 0.05$, $**P < 0.01$. (c) Expected calibration error across splits. (d) Per-pathway LR-Lex AUROC against the fraction of test-gene occurrences seen in training; marker area is proportional to test pairs. Dashed lines mark chance.

TABLE VI
ENSEMBLING, PRECISION–RECALL, CONFORMAL COVERAGE AND CURATION RANKING (TASK M). AUPRC: MACRO AUPRC; COV.: EMPIRICAL COVERAGE OF 90% SPLIT-CONFORMAL SETS (LR-LEX); P@10 LIFT: PRECISION@10 DIVIDED BY PREVALENCE (LR-LEX). PATHWAY VALUES ARE TEST-SIZE-WEIGHTED.

	Pair	Gene	Family	Pathway
Ensemble AUROC	0.988	0.924	0.814	0.659
LR-Lex AUROC	0.978	0.939	0.811	0.683
LR-Lex AUPRC	0.962	0.909	0.753	0.645
GBDT-Graph AUPRC	0.978	0.811	0.666	0.596
Ensemble AUPRC	0.980	0.882	0.740	0.630
Prevalence	0.400	0.399	0.401	0.426
Cov. (target 0.90)	0.906	0.851	0.728	0.580
P@10 lift, activation	1.80	1.79	1.66	1.16
P@10 lift, inhibition	2.81	2.87	2.77	1.30
P@10 lift, phosphorylation	3.34	3.44	3.24	2.13
Permuted-label AUROC	0.509	0.491	0.519	0.506

under every split (Fig. 7d), so the signal reported above is genuine, and the losses come from the absence of transferable knowledge rather than from errors in the evaluation pipeline.

VI. DISCUSSION

A. Implications for AI and LLM Knowledge Mining

Our results do not show that LLMs lack pathway knowledge; we did not evaluate LLMs. They show that the benchmark, as commonly used, cannot establish that they have it. Under random-row splitting a symbol-only classifier scores in the range attributed to LLM knowledge or prompt design, and the score is simultaneously inflated by pair overlap and deflated by co-annotation. An LLM is not trained on the split, but it was pre-trained on text that names the same gene families, so it has access to the same lexical regularities. Two observations make this concrete: within our own experiments, adding capacity (CharCNN, GNN) helped only where shortcuts existed, and the label that transfers across pathways is exactly the one whose cue is in the gene name. Prompt-optimization studies face an additional risk, because prompts selected on a validation set drawn from the same random split can overfit to pair- and family-level regularities [18].

B. Recommendations

We recommend that studies using KEGG-derived relation benchmarks to evaluate AI systems:

1) deduplicate to unique directed pairs and treat the task as multi-label, or report sign and mechanism as separate sub-tasks;
2) report family-disjoint and pathway-held-out results alongside any random split, together with the fraction of test pairs whose genes or families were seen;
3) include a symbol-only baseline (e.g., LR-Lex), a scrambled-symbol control and a node-profile baseline (Lookup) as floors that an AI system must beat;
4) report calibration (ECE or Brier score) and, where uncertainty is claimed, conformal coverage under each split, because confidence on unseen pathways is unreliable;
5) report per-label metrics, because pooled scores are dominated by the comparatively easy phosphorylation label;
6) select prompts or hyperparameters on validation data that are disjoint from the test set at the same level as the claim being made.

All splits in this study are generated deterministically from seeds in the released code so that future LLM evaluations can use identical partitions.

C. Limitations

We analysed one public benchmark of 23 human pathways; other KEGG extractions, species or relation vocabularies may differ, although the mechanisms identified here are properties of KEGG annotation and HGNC naming rather than of this extraction. Our family definition is a suffix rule that groups most paralogs but misses irregularly named families; HGNC family assignments [20] would be more precise. We excluded 1.3% of records with rare relation types. Neural hyperparameters were fixed rather than tuned, which avoids test leakage but may understate their best attainable performance. Most

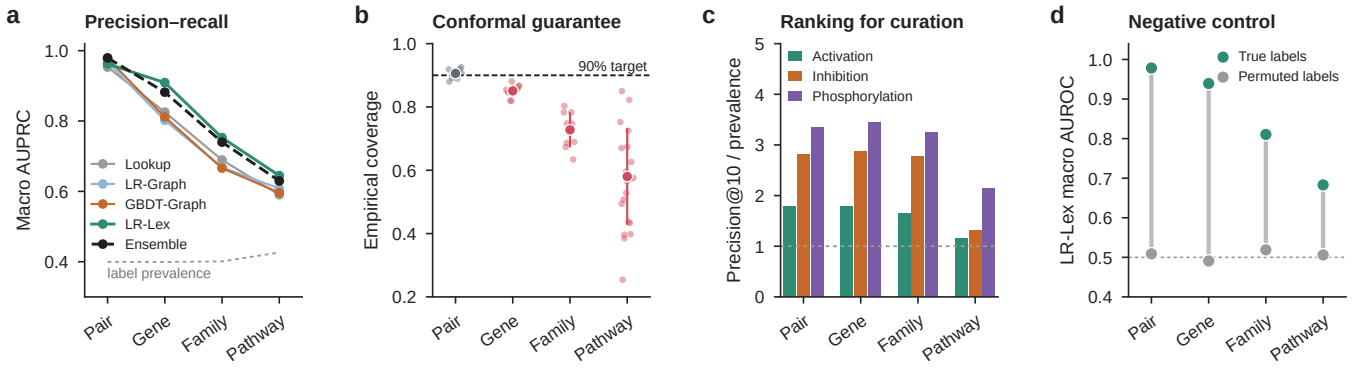


Fig. 7. Ensembling, uncertainty and ranking. (a) Macro AUPRC across splits for four models and their ensemble; the dashed grey line is label prevalence. (b) Empirical coverage of split-conformal prediction sets at a nominal 90% level (dashed line); small points are folds or held-out pathways, large points are means with one standard deviation. (c) Precision among the ten top-ranked test pairs per label, divided by prevalence; the dashed line marks random ranking. (d) LR-Lex AUROC with true and with permuted training labels; the dashed line marks chance.

importantly, we did not run LLMs, so we cannot quantify how much of a particular model's reported score comes from the shortcuts described here; evaluating LLMs under the proposed splits is the most important next step.

VII. Conclusion

High scores on KEGG-derived relation benchmarks can be obtained from gene names alone. Once label co-annotation, pair overlap and gene-family naming are controlled, every model we tested, including a character-level CNN, a graph neural network and their ensemble, falls toward chance when predicting whether a new interaction activates or inhibits its target in an unseen pathway; only phosphorylation, which kinase names reveal, transfers. Stress tests confirm that the models rely on name spelling and gene roles and become overconfident where they fail, with split-conformal coverage dropping from 91% to 58% on held-out pathways. AI and LLM knowledge mining should therefore be evaluated per label under family- and pathway-disjoint splits, with calibration and symbol-only baselines. Future work will use the released splits to evaluate LLMs, test whether biological evidence beyond gene names, such as sequence, structure or literature, restores transfer of regulatory sign, and extend the benchmark to other pathway resources and species.

Data and Code Availability

The combined relation table, split definitions, analysis code, figure scripts and per-fold results generated in this study are available at https://github.com/Muh-aza/aaa.

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